Circles

Class 4

Note for the teacher.

- **Aims.** Students decide when four points lie on one circle. They explain why a segment that subtends equal angles at two points on the same side gives four concyclic points, why opposite angles of a cyclic quadrilateral add up to 180°, and the converse. They use these results to find angles and to recognise cyclic quadrilaterals.
- **NCERT links.** Theorems 10, 11 and 12 keep their NCERT numbers. The questions include the NCERT in-text exercise in Section 5.8 and End-of-Chapter Questions 7, 8, 10, 14, 15 and 21.
- **Prior work.** Class 3: Theorem 9, angles in the same segment, the angle in a semicircle, and P7 (a point inside the circle gives a larger angle, a point outside a smaller one).
- **Materials.** One A4 sheet and the geometry box for each student.
- **Method.** Theorems 10 and 12 are proved by contradiction. P3 and P6 lead students through each argument one step at a time; let them write each step before it is discussed.

1 When do four points lie on a circle?

Points that lie on one circle are called **concyclic**. Any three points that are not on one line are concyclic (Theorem 5). What about four points?

Solve first.

P1. Draw a rough figure for each part.

(a) $ABCD$ is a rectangle with diagonals meeting at O. Explain why $OA = OB = OC = OD$. Are A, B, C and D concyclic?

(b) Draw a rhombus $PQRS$ with side 4 cm and $\angle P = 60°$. Draw the circle through P, Q and S. Does it pass through R?

Answers.

P1.

(a) The diagonals of a rectangle are equal and bisect each other, so $OA = OB = OC = OD$. The circle with centre O and radius OA passes through all four vertices: they are concyclic.

FIG

(b) No. $\triangle PQS$ is equilateral, and its circumcircle has radius about 2.3 cm. R lies outside this circle. So four points need not be concyclic.

2 Equal angles on the same side

Solve first.

P2. A, B, C are points on a circle. D is a point on the same side of AB as C, and $\angle ADB = \angle ACB$. Suppose D is *outside* the circle. Then AD cuts the circle at a point E between A and D.

(a) Why is $\angle AEB = \angle ACB$?

(b) $\angle AEB$ is an exterior angle of $\triangle BED$. Which is larger, $\angle AEB$ or $\angle ADB$?

(c) Use (a), (b) and $\angle ADB = \angle ACB$. What goes wrong?

(d) What goes wrong if D is *inside* the circle? (Extend AD to meet the circle at E.)

(e) So where must D be?

Answers.

P2.

(a) C and E are in the same segment of chord AB, so the angles are equal (Class 3).

FIG

(b) An exterior angle is greater than each interior opposite angle, so $\angle AEB > \angle ADB$.

(c) Then $\angle ACB = \angle AEB > \angle ADB = \angle ACB$: an angle greater than itself. This is impossible.

(d) Now $\angle ADB$ is the exterior angle of $\triangle BDE$, so $\angle ADB > \angle AEB = \angle ACB$. This is also impossible.

(e) On the circle through A, B and C.

Theorem 10 (NCERT Theorem 10). *If a line segment AB joining two points A, B subtends equal angles at two other points C, D that lie on the same side of AB, then the four points lie on a circle.*

FIG

Why it works.

Given: C and D lie on the same side of AB, and $\angle ACB = \angle ADB$.
To show: A, B, C and D lie on one circle.

1. A, B and C are not on one line, so one circle passes through them (Theorem 5).
2. If D were outside this circle, P2(c) shows that $\angle ACB > \angle ACB$. If D were inside, P2(d) gives the same kind of contradiction.
3. So D lies on the circle.

Theorem 10 is the converse of "angles in the same segment are equal".

Example 1. C and D lie on the same side of AB, with $\angle ACB = \angle ADB = 40°$ and $\angle CAD = 25°$. Find $\angle CBD$.

FIG

Given: $\angle ACB = \angle ADB = 40°$, C and D on the same side of AB, $\angle CAD = 25°$.
To find: $\angle CBD$.

1. A, B, C and D are concyclic (Theorem 10).
2. $\angle CBD$ and $\angle CAD$ stand on the chord CD, in the same segment. So $\angle CBD = \angle CAD = 25°$.

Answer: $\angle CBD = 25°$.

3 Cyclic quadrilaterals

A quadrilateral whose four vertices lie on a circle is a **cyclic quadrilateral**.

Solve first.

P3. $ABCD$ is a cyclic quadrilateral in a circle with centre O. The arc BCD subtends $\angle BOD = 140°$ at O.

(a) A is outside the arc BCD. Find $\angle BAD$.

(b) What angle does the arc BAD subtend at O? Find $\angle BCD$.

(c) Find $\angle BAD + \angle BCD$.

P4. Repeat P3 with $\angle BOD = 100°$. What do you notice?

P5. In P3, find $\angle ABC + \angle ADC$ without any more information.

Answers.

P3.

(a) $\angle BAD = \dfrac{140°}{2} = 70°$ (Theorem 9).

FIG

(b) The reflex angle $360° - 140° = 220°$. C is outside the arc BAD, so $\angle BCD = \dfrac{220°}{2} = 110°$.

(c) $70° + 110° = 180°$.

P4. $\angle BAD = 50°$ and $\angle BCD = \dfrac{260°}{2} = 130°$. The sum is again 180°: the two arcs together make the whole circle, 360°, and each angle is half of its arc's angle.

P5. The angles of a quadrilateral add up to 360°. So $\angle ABC + \angle ADC = 360° - 180° = 180°$.

Theorem 11 (NCERT Theorem 11). *The sum of two opposite angles of a cyclic quadrilateral is* 180°.

FIG

Why it works.

Given: $ABCD$ is a cyclic quadrilateral in a circle with centre O.
To show: $\angle BAD + \angle BCD = 180°$.

1. The arc BCD subtends an angle p at O, and the arc BAD subtends the rest of the complete angle, $q = 360° - p$.
2. A is outside the arc BCD, so $\angle BAD = \dfrac{p}{2}$. C is outside the arc BAD, so $\angle BCD = \dfrac{q}{2}$ (Theorem 9).
3. $\angle BAD + \angle BCD = \dfrac{p+q}{2} = \dfrac{360°}{2} = 180°$.

The other pair also adds up to 180°, because the four angles add up to 360° (P5).

Solve first.

P6. $ABCD$ is a quadrilateral with $\angle BAD + \angle BCD = 180°$. Draw the circle through A, B and D. Suppose C is *outside* this circle, and CD cuts the circle at E, between D and C.

(a) $ABED$ is a cyclic quadrilateral. Find $\angle BAD + \angle BED$.

(b) Compare $\angle BED$ with $\angle BCD$.

(c) $\angle BED$ is an exterior angle of $\triangle BEC$. What goes wrong?

(d) So where must C be?

Answers.

P6.

(a) 180° (Theorem 11).

(b) Both $\angle BED$ and $\angle BCD$ make 180° with $\angle BAD$, so $\angle BED = \angle BCD$.

(c) An exterior angle is greater than each interior opposite angle, so $\angle BED > \angle BCE = \angle BCD$. This contradicts (b).

(d) C cannot be outside the circle. Exercise 4A Question 10 shows that it cannot be inside either. So C is on the circle.

FIG

Theorem 12 (NCERT Theorem 12). *If two opposite angles of a quadrilateral add up to* 180°*, then the vertices of the quadrilateral lie on a circle, that is, they are concyclic.*

Why it works.

Given: quadrilateral $ABCD$ with $\angle BAD + \angle BCD = 180°$.
To show: $ABCD$ is cyclic.

1. Draw the circle through A, B and D (Theorem 5).
2. If C were outside it, P6 gives $\angle BED = \angle BCD$ and $\angle BED > \angle BCD$ together, which is impossible. The case of C inside is similar.
3. So C lies on the circle, and $ABCD$ is cyclic.

Things to Remember

Opposite angles of a cyclic quadrilateral add up to 180°. If opposite angles of a quadrilateral add up to 180°, it is cyclic.
If a segment subtends equal angles at two points on the same side of it, the four points are concyclic.

Example 2. $ABCD$ is a cyclic quadrilateral. AB is a diameter of the circle, and $\angle ADC = 120°$. Find $\angle BAC$.

FIG

Given: $ABCD$ cyclic, AB a diameter, $\angle ADC = 120°$.
To find: $\angle BAC$.

1. $\angle ABC = 180^\circ - 120^\circ = 60^\circ$ (opposite angles of a cyclic quadrilateral, Theorem 11).
2. $\angle ACB = 90^\circ$ (angle in a semicircle).
3. In $\triangle ABC$, $\angle BAC = 180^\circ - 90^\circ - 60^\circ = 30^\circ$.

Answer: $\angle BAC = 30^\circ$.

Draw a rough figure for every question in Exercise 4A.

Exercise 4A

1. $ABCD$ is a cyclic quadrilateral. If $\angle A = 75^\circ$, find $\angle C$. If $\angle B = 110^\circ$, find $\angle D$.
2. $PQRS$ is a cyclic quadrilateral with $\angle P = (2x + 10)^\circ$ and $\angle R = (3x - 20)^\circ$. Find x, $\angle P$ and $\angle R$.
3. A quadrilateral $ABCD$ has $\angle A = 80^\circ$, $\angle B = 110^\circ$, $\angle C = 100^\circ$ and $\angle D = 70^\circ$. Can it be cyclic? Explain.
4. Shreya says, "Quadrilateral $ABCD$ has $\angle A = 100^\circ$, $\angle B = 80^\circ$, $\angle C = 100^\circ$ and $\angle D = 80^\circ$. Its angles 100° and 80° add up to 180°, so it is cyclic." Find her mistake.
5. Four trees P, Q, R and S stand at the corners of a quadrilateral plot, with $\angle P = 75^\circ$, $\angle Q = 95^\circ$ and $\angle R = 105^\circ$. A gardener wants to lay a circular path through all four trees. Is it possible? Explain.
6. Show that a rectangle is the only parallelogram that can be cyclic.
7. A rectangle is inscribed in a circle. Show that its diagonals meet at the centre of the circle.
8. $ABCD$ is a cyclic quadrilateral, and E is a point on CD extended beyond D. Explain why $\angle ADE = \angle ABC$.
9. $ABCD$ is a trapezium with $AB \parallel DC$ and $AD = BC$. Show that $ABCD$ is cyclic.
10. In P6, suppose C is *inside* the circle through A, B and D. Extend DC to meet the circle at E. Show that this is also impossible.
11. A cyclic quadrilateral has sides 5, 5, 12 and 12 units. Find its area.
12. Draw a circle of radius 4 cm and mark four points on it to form a quadrilateral. Measure its four angles and check Theorem 11. Then draw a quadrilateral whose vertices are *not* all on one circle, and check whether its opposite angles add up to 180°.

Note for the teacher. Question 4 must be discussed in class: Shreya added two adjacent angles, not opposite ones. Questions 6, 7 and 9 together show which special quadrilaterals are cyclic. If time is short, set Questions 10, 11 and 12 as homework.

Answers.

1. $\angle C = 180° - 75° = 105°$ and $\angle D = 180° - 110° = 70°$ (Theorem 11).

2. P and R are opposite angles, so $(2x + 10) + (3x - 20) = 180$. Then $5x - 10 = 180$, so $x = 38$. $\angle P = 86°$ and $\angle R = 94°$.

3. Yes. $\angle A + \angle C = 80° + 100° = 180°$ (and $\angle B + \angle D = 180°$), so it is cyclic by Theorem 12.

4.

FIG

$100°$ and $80°$ are *adjacent* angles ($\angle A$ and $\angle B$). The opposite angles are $\angle A$ and $\angle C$: $100° + 100° = 200° \neq 180°$. So $ABCD$ is not cyclic. (It is a parallelogram that is not a rectangle.)

5. $\angle S = 360° - 75° - 95° - 105° = 85°$. $\angle P + \angle R = 75° + 105° = 180°$, so the plot is cyclic (Theorem 12). Yes, one circular path passes through all four trees.

6.

FIG

1. Opposite angles of a parallelogram are equal: $\angle A = \angle C$.
2. If it is cyclic, $\angle A + \angle C = 180°$ (Theorem 11), so $\angle A = \angle C = 90°$.
3. A parallelogram with a right angle is a rectangle.

So the only cyclic parallelograms are rectangles. (This also settles P1(b): a rhombus that is not a square is not cyclic.)

7.

FIG

1. $\angle ABC = 90°$. B is outside the arc AC that does not contain B, so this arc subtends $2 \times 90° = 180°$ at the centre O (Theorem 9).
2. So A, O and C are on one line: AC is a diameter, and O is its midpoint. In the same way, BD is a diameter with midpoint O.
3. So the diagonals meet at O.

8.

FIG

$\angle ADE + \angle ADC = 180°$ (linear pair) and $\angle ABC + \angle ADC = 180°$ (Theorem 11). So $\angle ADE = \angle ABC$: the exterior angle equals the interior opposite angle.

9.

FIG

1. Draw $DP \perp AB$ and $CQ \perp AB$. $\triangle APD \cong \triangle BQC$ (RHS: $AD = BC$, $DP = CQ$), so $\angle A = \angle B$.
2. $AB \parallel DC$, so $\angle A + \angle D = 180°$ (co-interior angles).
3. So $\angle B + \angle D = 180°$, and $ABCD$ is cyclic (Theorem 12).

10.

1. $ABED$ is cyclic, so $\angle BAD + \angle BED = 180°$. Also $\angle BAD + \angle BCD = 180°$. So $\angle BED = \angle BCD$.
2. C is between D and E, so $\angle BCD$ is an exterior angle of $\triangle BCE$. Then $\angle BCD > \angle BEC = \angle BED$.
3. The two statements contradict each other. So C cannot be inside the circle.

FIG

11.

1. If the sides are in the order 5, 12, 5, 12, opposite sides are equal, so it is a parallelogram. A cyclic parallelogram is a rectangle (Question 6). Area $= 5 \times 12 = 60$ square units.
2. If the order is 5, 5, 12, 12, it is a kite $ABCD$ with $AB = AD = 5$ and $CB = CD = 12$. $\triangle ABC \cong \triangle ADC$ (SSS), so $\angle B = \angle D$. By Theorem 11, $\angle B + \angle D = 180°$, so each is 90°. Area $= 2 \times \frac{1}{2} \times 5 \times 12 =$ 60 square units.

FIG

Answer: 60 square units in both cases.

12. For the cyclic quadrilateral, each pair of opposite angles adds up to about 180° (within 2° for measuring). For the other quadrilateral, the sums differ from 180°.

4 Further questions

F1. $ABCD$ is a cyclic quadrilateral whose diagonals meet at P. $\angle DBC = 35°$ and $\angle BAC = 50°$. Find $\angle BAD$ and $\angle BCD$.

F2. ABC is a triangle, and BE and CF are its altitudes. Show that B, C, E and F are concyclic.

Answers.

F1.

1. $\angle DAC = \angle DBC = 35°$ (same segment, chord DC).
2. $\angle BAD = \angle BAC + \angle CAD = 50° + 35° = 85°$.
3. $\angle BCD = 180° - 85° = 95°$ (Theorem 11).

FIG

Answer: $\angle BAD = 85°$, $\angle BCD = 95°$.

F2.

$\angle BEC = \angle BFC = 90°$, and E and F lie on the same side of BC. So BC subtends equal angles at E and F, and B, C, E, F are concyclic (Theorem 10). The circle has BC as a diameter.

FIG

5 The twelve theorems of the chapter

Class	Our no.	NCERT no.	Statement
1	1	2	Equal chords subtend equal angles at the centre.
1	2	3	Chords that subtend equal angles at the centre are equal.
1	3	4	The line from the centre to the midpoint of a chord is perpendicular to the chord.
1	4	5	The perpendicular from the centre to a chord bisects the chord.
2	5	1	There is a unique circle through three non-collinear points.
2	6	6	Equal chords are equidistant from the centre.
2	7	7	Chords equidistant from the centre are equal.
2	8	8	Of two chords, the longer one is nearer the centre.
3	9	9	The angle at the centre is double the angle at a point on the rest of the circle. *Corollary:* the angle in a semicircle is 90°.
4	10	10	If AB subtends equal angles at C and D on the same side, then A, B, C, D are concyclic.
4	11	11	Opposite angles of a cyclic quadrilateral add up to 180°.
4	12	12	If opposite angles of a quadrilateral add up to 180°, it is cyclic.